

Apache Iceberg

Creating tables

Lesson objectives

Apache Iceberg: Creating tables

1. Explain the syntax used to create Iceberg tables.
2. Demonstrate the syntax used to control table properties, including file format and partition strategy.
3. Explain the nature of Iceberg architecture, including the use of manifest files, and manifest lists.
4. Explain the role that metadata columns and tables play in Iceberg.
5. Demonstrate how to create and utilize Iceberg tables.

CREATE (an Iceberg) TABLE

Starburst Enterprise

Use a schema located in a catalog leveraging the [Iceberg](#) connector. Example is using all default values.

```
CREATE TABLE customer (  
    name varchar, address varchar)
```

Starburst Galaxy

Use a schema located in a catalog leveraging the [Great Lakes connector](#). Example is using default values, but explicitly identifying table format.

```
CREATE TABLE customer (  
    name varchar, address varchar)  
  
WITH (type='iceberg');
```

CREATE TABLE AS (CTAS) ALSO WORKS AS EXPECTED

Iceberg table property: `type`

Use this property in Starburst Galaxy to override the default table type of the catalog.

Example:

```
CREATE TABLE customer (  
    name varchar, address varchar)  
WITH (type='iceberg');
```

Iceberg table property: `format`

Optionally specifies the format of the table data files.

Valid values:

ORC	Apache ORC file format
PARQUET	Default value; Apache Parquet file format
AVRO	Apache Avro file format

Example:

```
CREATE TABLE customer (  
    name varchar, address varchar)  
  
WITH (type='iceberg',  
      format='orc');
```

Iceberg table property: partitioning

Optionally specifies table partitioning strategy.

Valid values:

```
ARRAY [ 'c1', 'c2' ]
```

A comma-separated list of the partitioned-by columns are placed within the square-brackets. *In this example, a table would be partitioned by columns **c1** and **c2**.*

No default value.

Partition column(s) do NOT have to be listed last

Iceberg table property: `location`

Optionally specifies the file system location URI for the table.

Valid values:

A valid, reachable URI	Example: <code>s3://my-bucket/at/this/path</code>
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Defaults to the URI of the schema the table is being created into, postfixed with a derivative of the table name.

This does NOT make it an EXTERNAL table, just allows the folder location to be controlled

Iceberg table property: `format_version`

Optionally specifies the format version of the Iceberg specification.

Valid values:

1	Original version which was designed for analytical data tables
2	Adds row-level updates and deletes for analytical tables with immutable files

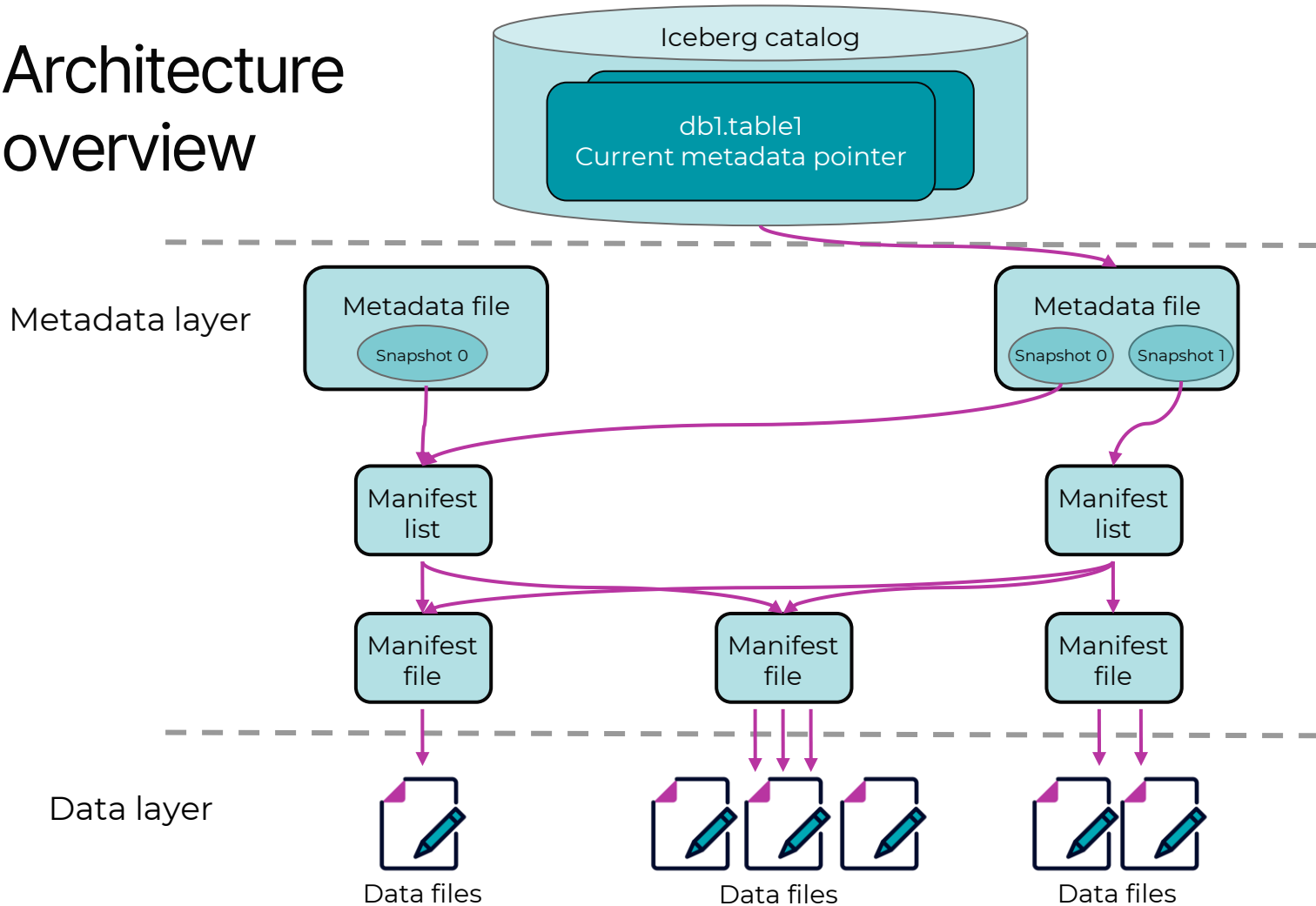
Defaults to 2.

Bloom filter properties

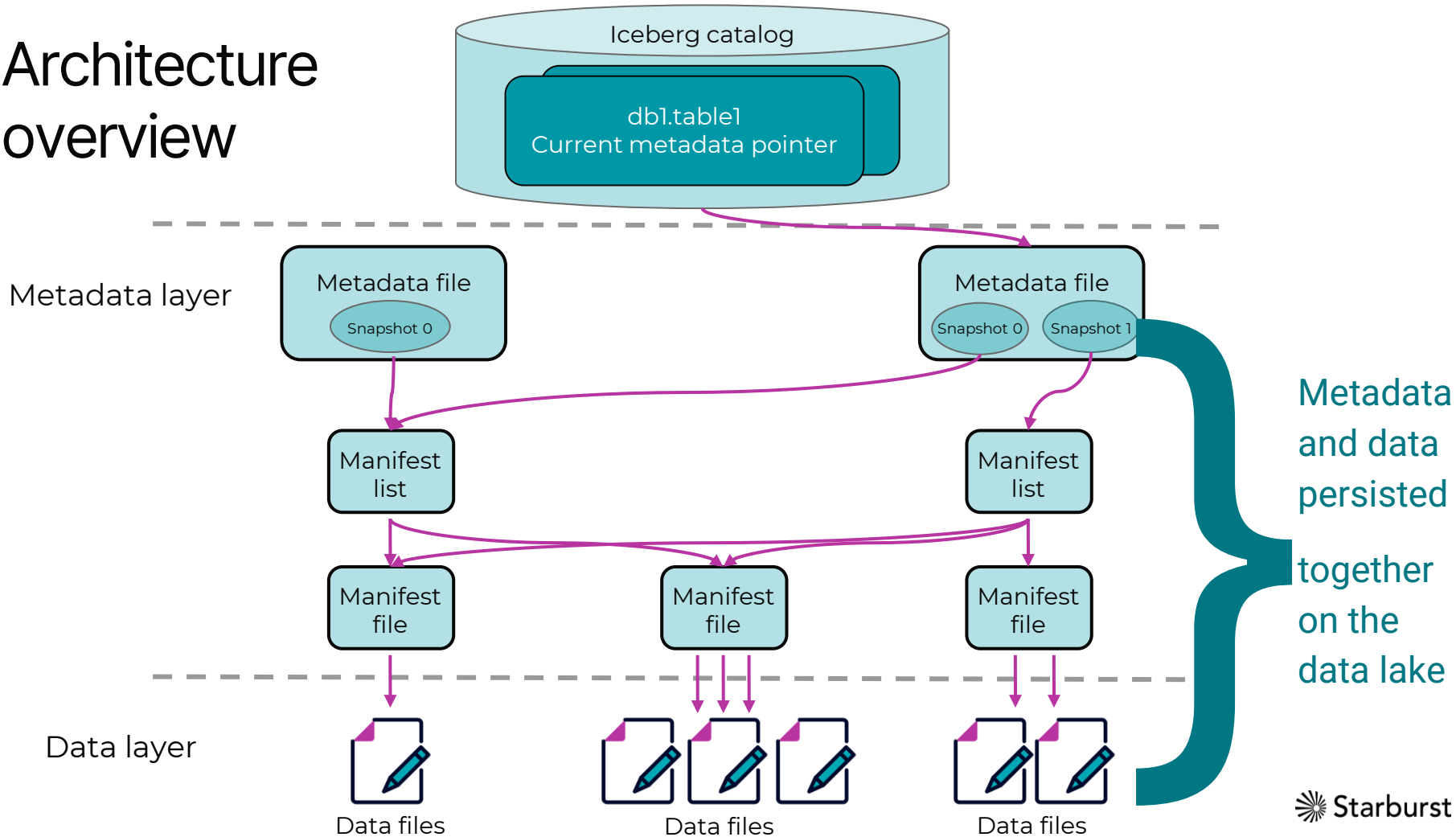
Additional optional properties for use with ORC bloom filter functionality. [ORC file format is required.](#)

PROPERTY	DESCRIPTION	DEFAULT
<code>orc_bloom_filter_columns</code>	Comma separated list of columns to use for ORC bloom filter. Improves query performance when using equality and IN predicates..	<code>[]</code>
<code>orc_bloom_filter_fpp</code>	The ORC bloom filters false positive probability.	<code>0.05</code>

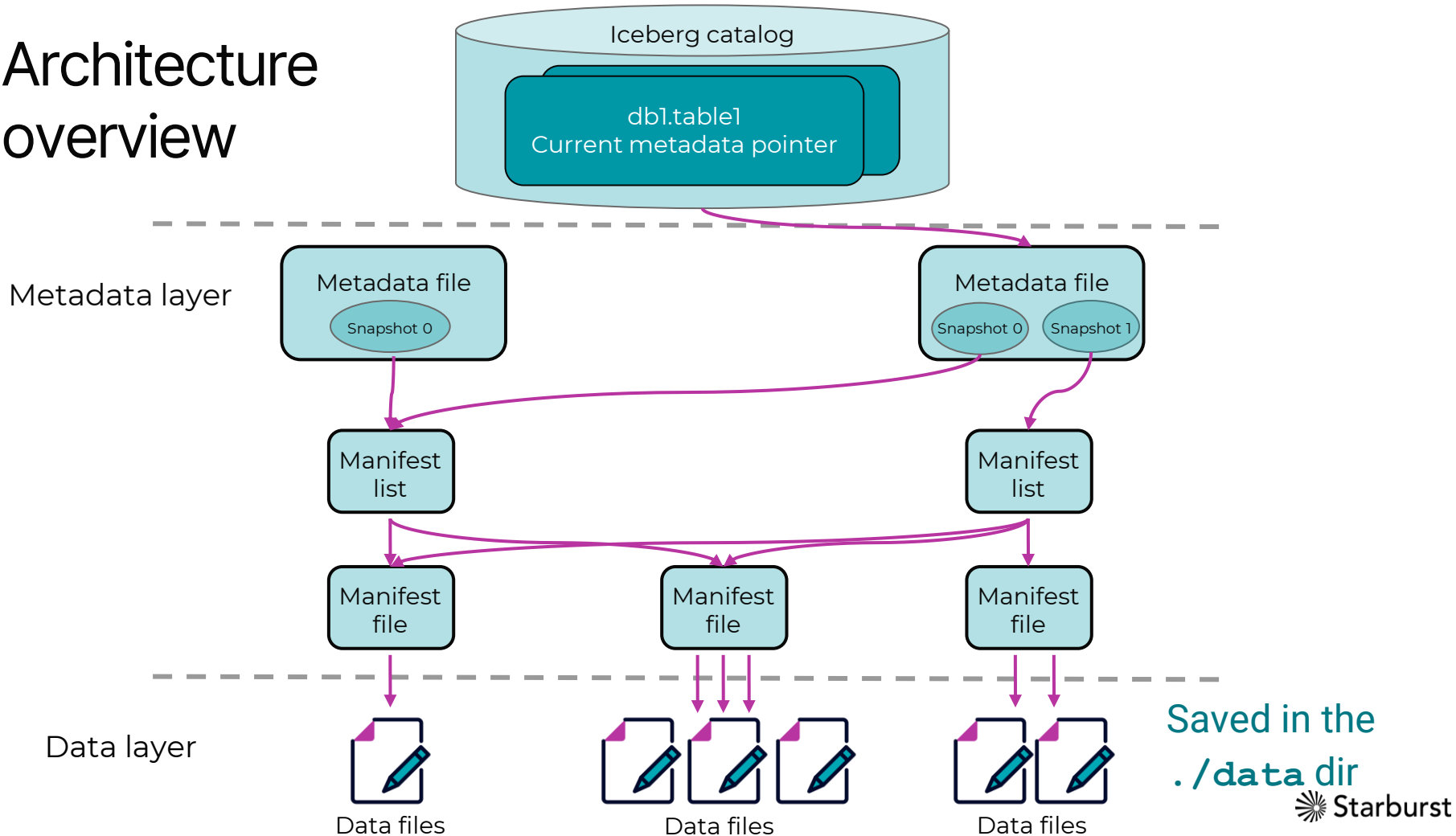
Architecture overview



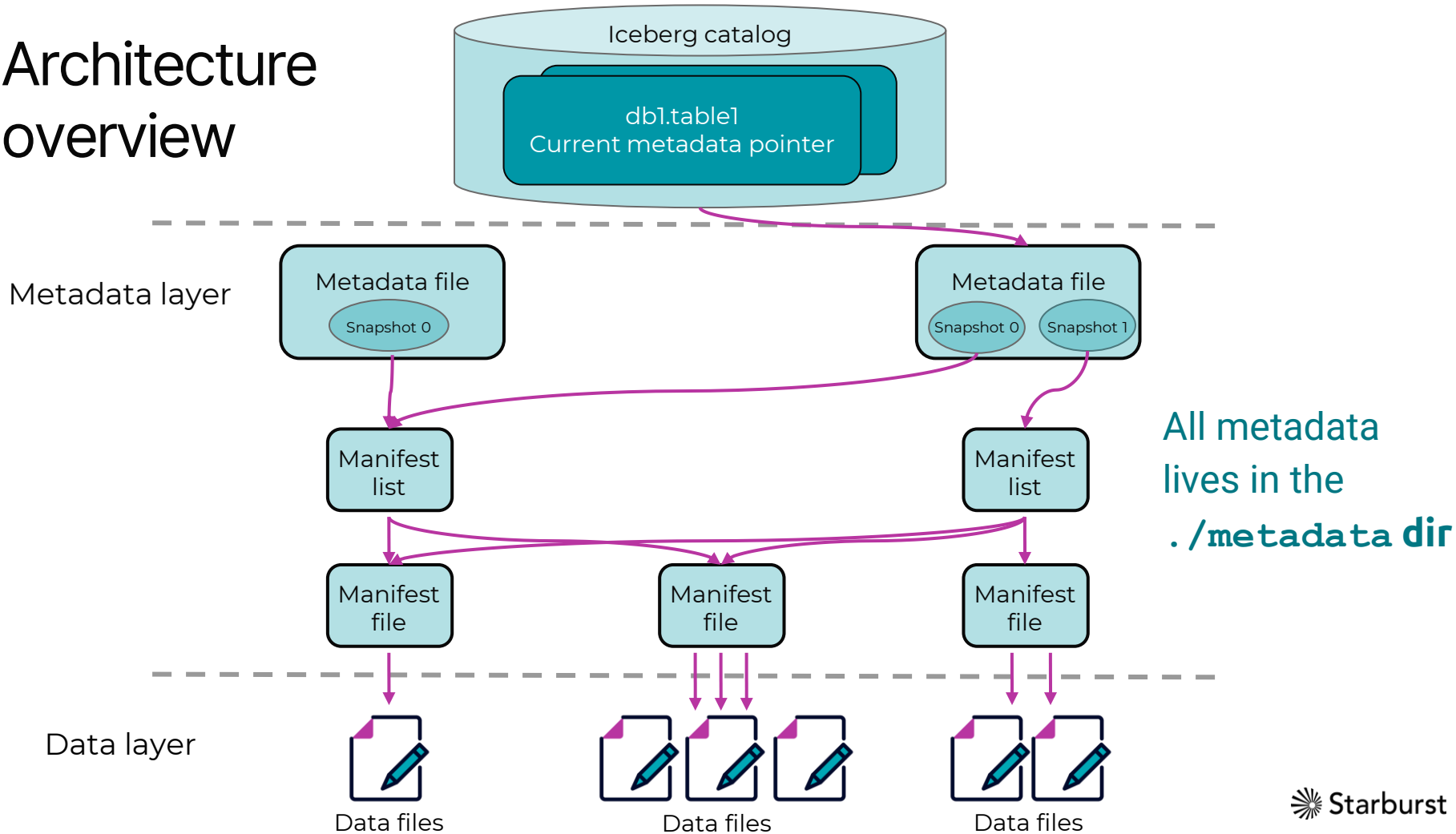
Architecture overview



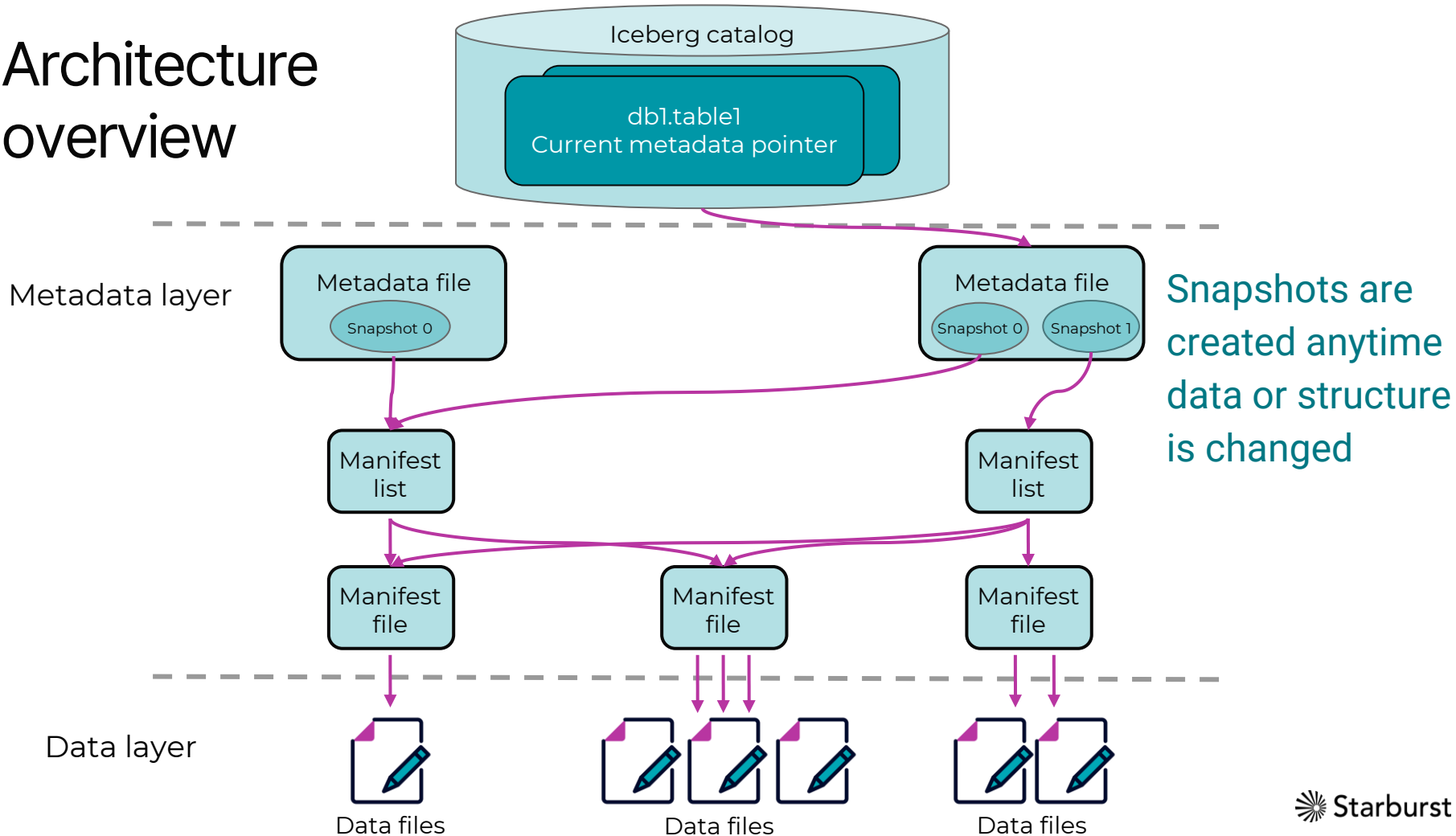
Architecture overview



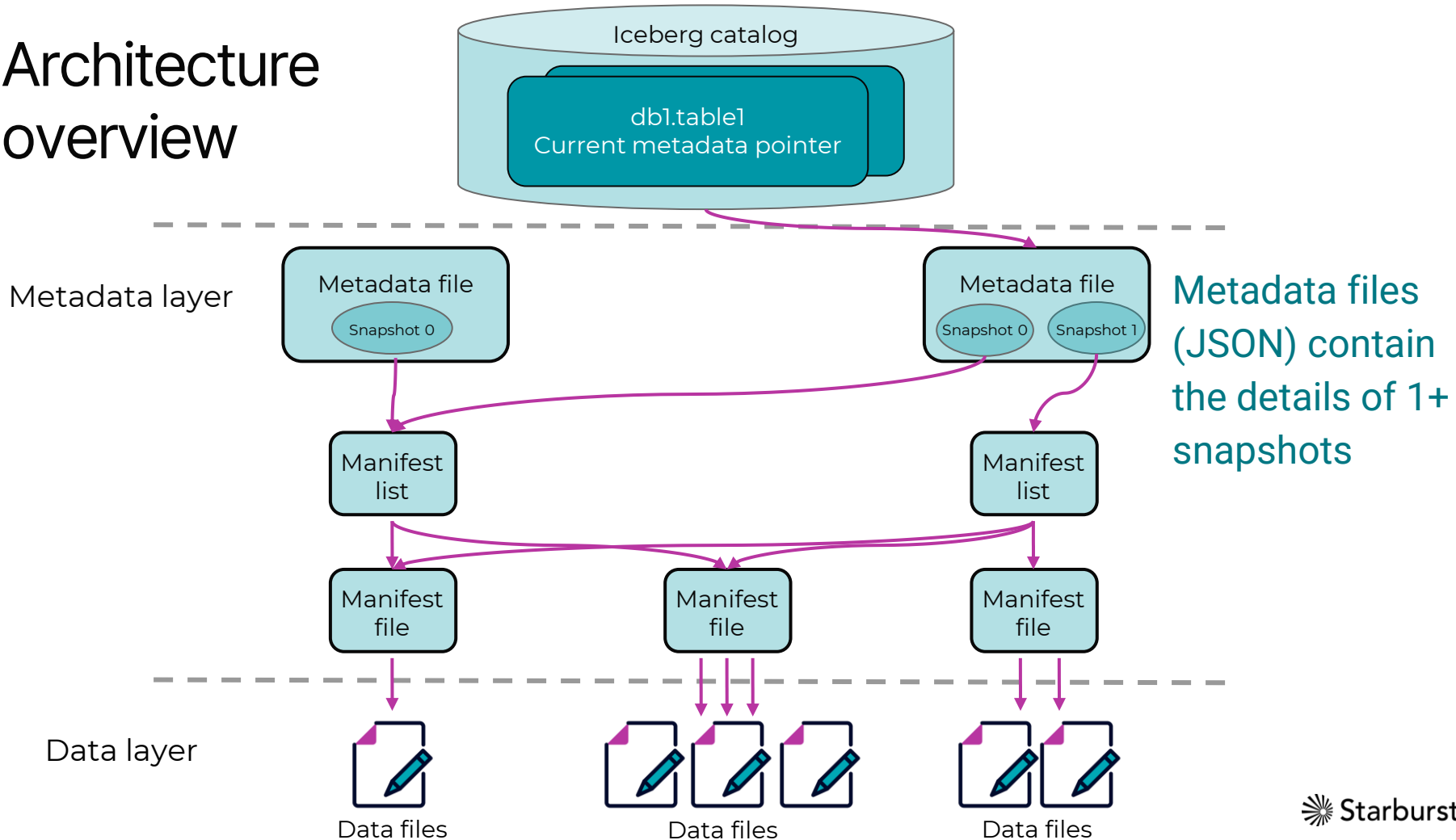
Architecture overview



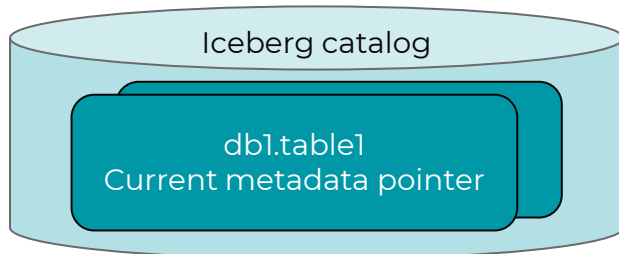
Architecture overview



Architecture overview

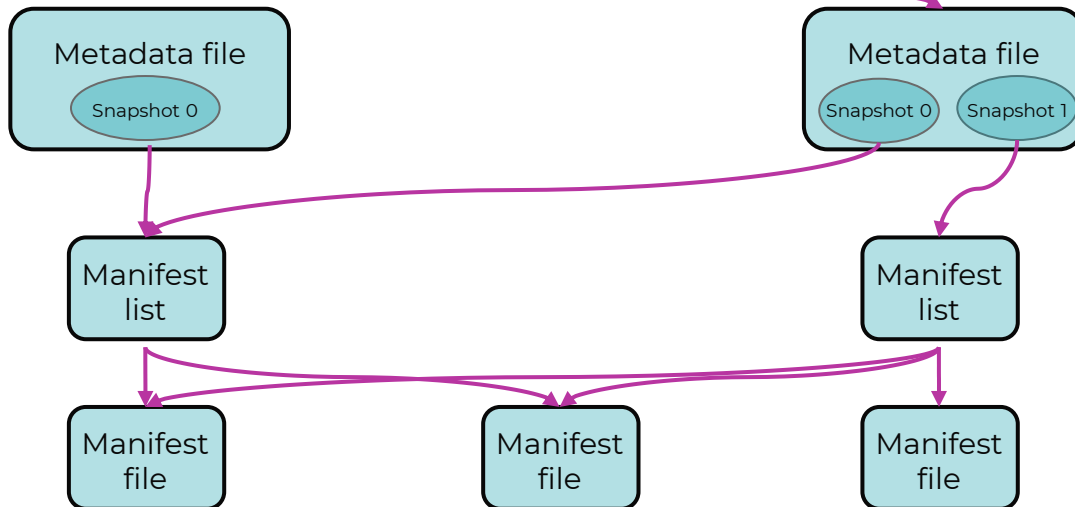


Architecture overview



A reference to the current snapshot is stored in HMS

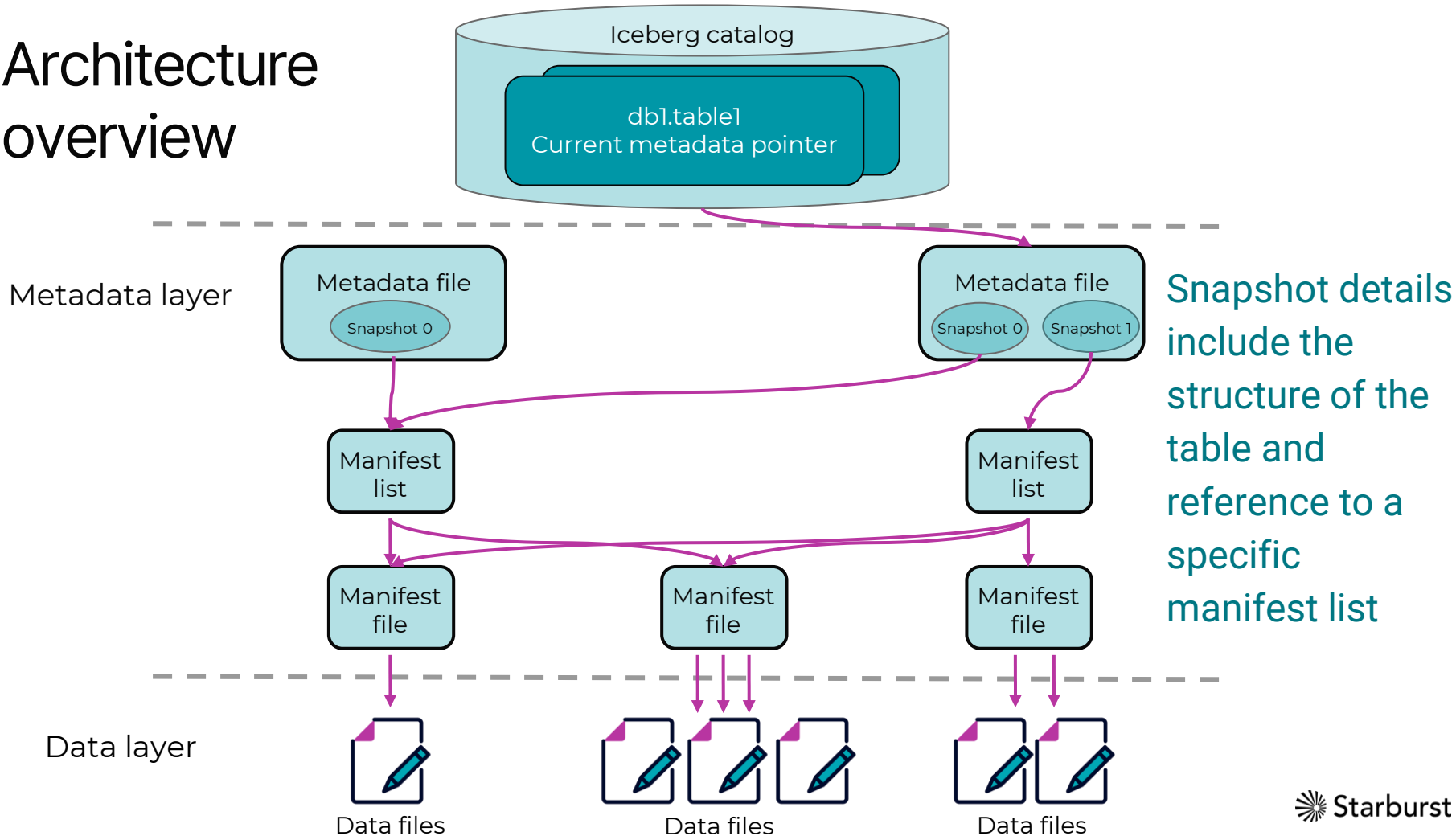
Metadata layer



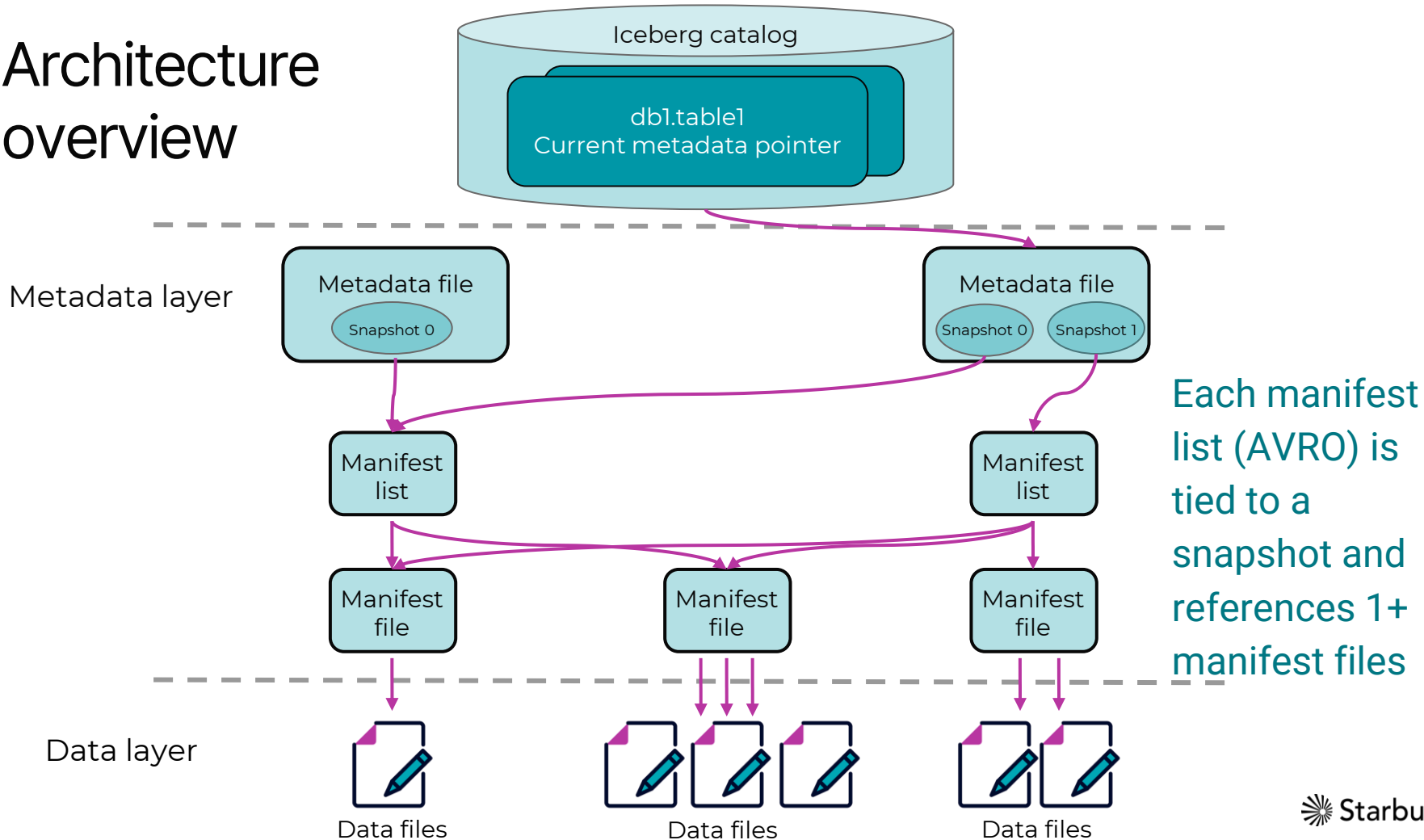
Data layer



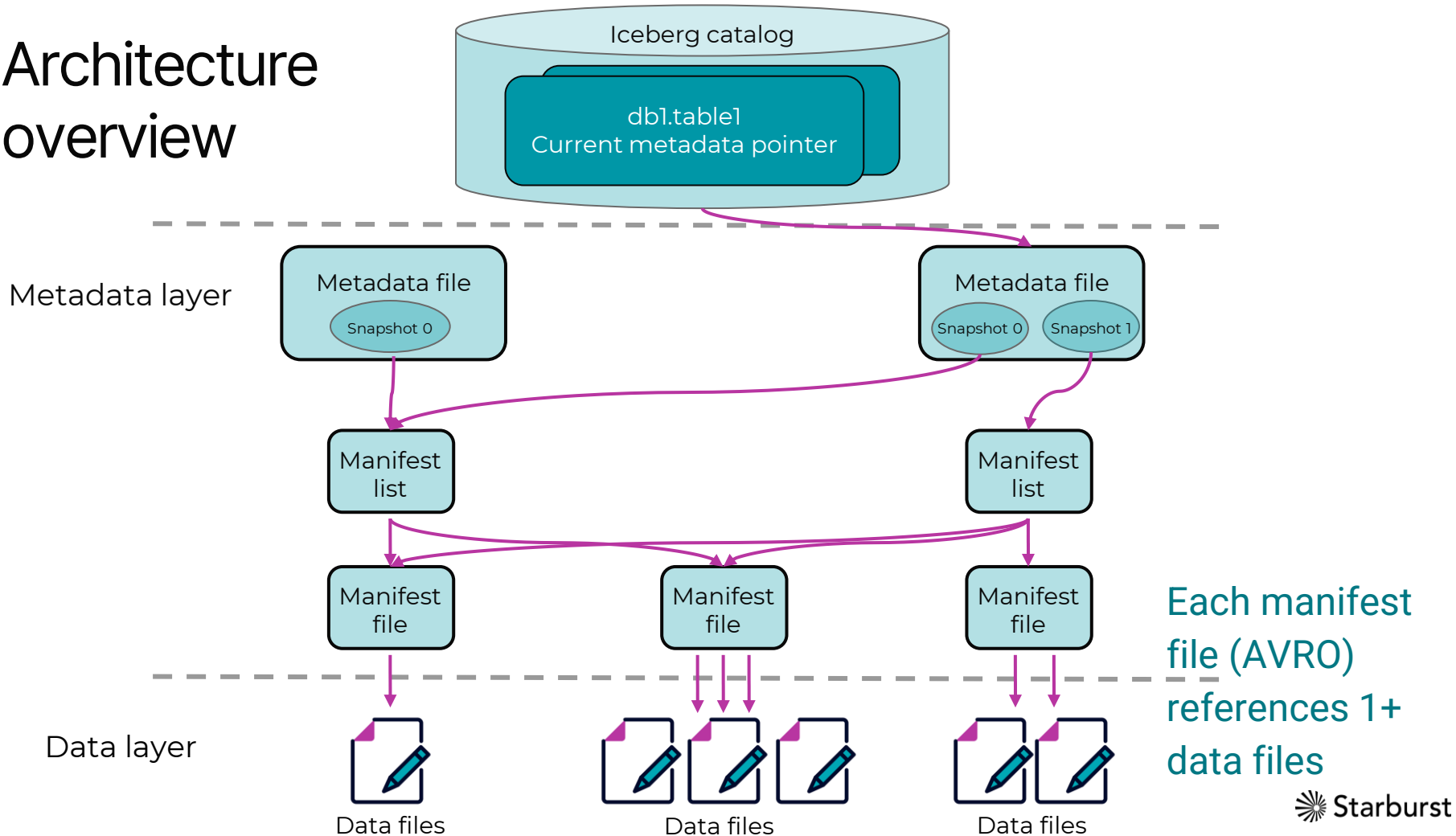
Architecture overview



Architecture overview



Architecture overview



Access to metadata

Metadata columns

Much like available from the original Hive table format, hidden columns are available on all tables.

- `$path` – full file system path name of the file for a given row
- `$file_modified_time` – timestamp of the last modification of the file for a given row

Metadata tables

Metatable tables, including the following, contain information about the internal structure of each Iceberg table

- `$history`
- `$snapshots`
- `$manifests`
- `$partitions`
- `$files`

Instructor demonstration

How Iceberg tables are stored (15 mins)



Hands-on exercises

Lab 1: Create and populate
Iceberg tables (30 mins)

Lab 2: Explore partitions with
Iceberg (30 mins)